

Honest Ranking: A Client-Grouped Audit of CTR / Engagement Opportunity Scoring

FlyRank ML Internship Capstone
Lane 4: CTR / Engagement Opportunity Scoring

July 21, 2026

Abstract

This report asks whether a transparent rule and a simple supervised model can help a content team honestly prioritize which search-landing pages to review first — and whether that help survives an honest validation check, not just a first look. Working within one scoring lane on an anonymized 30,000-row production sample across 32 clients, we built a hand-written baseline rule, trained a Logistic Regression model on a client-grouped split, and then re-audited both under a deliberately harder validation design. Logistic Regression showed a real average edge over the baseline (Precision@50: 0.50 → 0.74 on the primary split), but that edge held in only 3 of 5 differently-seeded honest splits, and a naive, non-grouped split would have overstated it further still (ROC AUC 0.72 vs. 0.61). A leakage audit found that nearly all of the model's signal comes from features that share a time window with the label rather than from durable content properties, so it is best read as detecting pages that already resemble other declining pages, not as forecasting decline from independent evidence. The deployed output is a ranked, reason-coded action playbook built on the transparent rule rather than the model, with explicit human-review rules and a no-go list for what should not be automated.

0.50 → 0.74		3 / 5	18,752	32
P@50, model	rule →	seeds where model won	eligible scored	pages anonymized clients

1 Introduction

Search content teams routinely ask a version of the same question: out of thousands of pages, which ones are worth an editor's time this week? A model score is not, by itself, an answer — it is an input that still needs a transparent reason, a plausible alternative explanation, and a human in the loop.

This report documents one attempt to build that input honestly, for one specific lane: pages that already rank reasonably well but under-convert that ranking into clicks or engagement — **CTR / Engagement Opportunity Scoring**. Rather than reaching straight for a model, the work follows a deliberately staged path: check the signals a rule would lean on before writing the rule; write the rule and read its own output skeptically; only then train a model and require it to beat the rule on identical, held-out data; then take a second, harder pass at that comparison — a proper audit, not just a scoreboard.

The result is not a claim that machine learning “solves” content prioritization. It is a specific, checked answer to a narrower question: under an honest validation design, how much does a model actually

add over a rule a person can read in thirty seconds — and what, precisely, should a team trust it to do unsupervised? (Answer, previewed: not much yet, and nothing unsupervised.)

2 Data

Release. The FlyRank ML Internship starter slice — a single pre-anonymized CSV bundled with the internship repository, distinct from the program’s larger warehouse release. The baseline rule and the model comparison were both built on this one slice, on purpose, so every comparison in this report is on the same rows.

Shape. One table, one row per page, 44 columns, covering **32 anonymized clients**. `content_id` and `client_id` are opaque hashes — no client names, domains, URLs, or query text appear anywhere in this table or in this report.

Time window. Every activity metric (impressions, clicks, sessions, engagement) is a **90-day trailing aggregate** as of the data snapshot. The trend label used throughout this report compares the most recent 30 days against the prior 30 — a sub-window of that same 90-day snapshot, not a separately observed future period.

Table 1: Row accounting for the analysis sample.

Step	Rows
Raw rows in the release	30,000
Excluded — no search visibility (0 impressions in window)	0
Excluded — too new to trust a 90-day aggregate (<90 days old)	0
Final analysis table	30,000

The bundled slice happened to already satisfy both exclusion rules for every row; they are applied and reported here in full regardless, since the next data pull may not.

3 Methodology

Assumptions, checked before use. Two, both audited empirically rather than assumed: (1) click-through rate varies meaningfully by ranking position — confirmed for four of five position tiers, but genuinely uncertain at the very top (`top_3` did not clearly out-perform `page_1` once low-volume noise was controlled); (2) CTR is only a trustworthy signal above a minimum visibility floor — confirmed, with CTR variance and the zero-click rate both dropping sharply above roughly 300 impressions in this sample.

Baseline rule. An `opportunity_score` blending three percentile-ranked components: how far a page’s CTR sits below its *own position tier’s empirical median* (not an assumed ranking, because of finding (1) above), its search visibility, and an engagement-rate gap — computed only for pages clearing the volume floor from finding (2). Every scored page gets exactly one reason code and one action label.

Label. A proxy target — whether a page’s short-term traffic trend reads as declining — not a confirmed real-world outcome. Never used as a model feature; used only as what the model predicts and what the rule is checked against.

Features. Twenty-six pre-decision features spanning visibility, position, engagement, and static content properties (word count, search volume, competition, content type, intent). Every feature that defines the label — the trend figure itself and the two 30-day windows behind it — is deliberately excluded from the feature list.

Model. Logistic Regression, selected over Random Forest after testing both: feature–label correlations were weak and roughly linear, and Random Forest did not out-perform Logistic Regression even after a hyperparameter sweep across tree depth, leaf size, estimator count, and class weighting — reported as a real result, not smoothed into “roughly comparable.”

Validation design. A client-grouped split, not a random row split. Pages from the same client share templates and editorial practices; a random split lets a model partially learn *which client a row belongs to* rather than genuinely generalizing. Section 4 quantifies exactly how much that distinction changes the answer.

Check 1: confirm the label-defining columns never appear as features — they don’t. **Check 2, subtler:** most numeric features are 90-day aggregates that partially overlap the label’s own 60-day window. A model trained only on genuinely static, non-overlapping features (word count, search volume, competition, content type, age/freshness tier) was tested in isolation to quantify how much signal depends on that overlap — see Section 4.

4 Results (vs. Baseline)

On the split used for the headline numbers, Logistic Regression leads clearly. Two further checks — a naive-split comparison and a five-seed stability run — both temper how much that lead should be trusted.

Table 2: Baseline vs. model, same client-grouped split, same metric.

Model	ROC AUC	P@20	P@50	P@100
Baseline (rule)	0.525	0.40	0.50	0.53
Logistic Regression	0.611	0.80	0.74	0.70

Check 1 — split honesty. Re-running the exact same model on a naive, non-grouped row split — the split a first attempt might reach for — produces a noticeably better-looking, less trustworthy number.

Check 2 — is the “after” number stable? One split is one draw of which 8 of 32 clients land in the test set. Re-running the honest method under five different seeds shows a real average edge for the model, but not a settled one.

The fair summary is “*a modest, directional average edge, not a settled win*” — not “*clearly beats the baseline.*” A model using only static, non-time-overlapping content features scored at chance (ROC AUC 0.51), confirming that almost all of the model’s edge traces back to features that share a time window with the label, not to independent content signal.

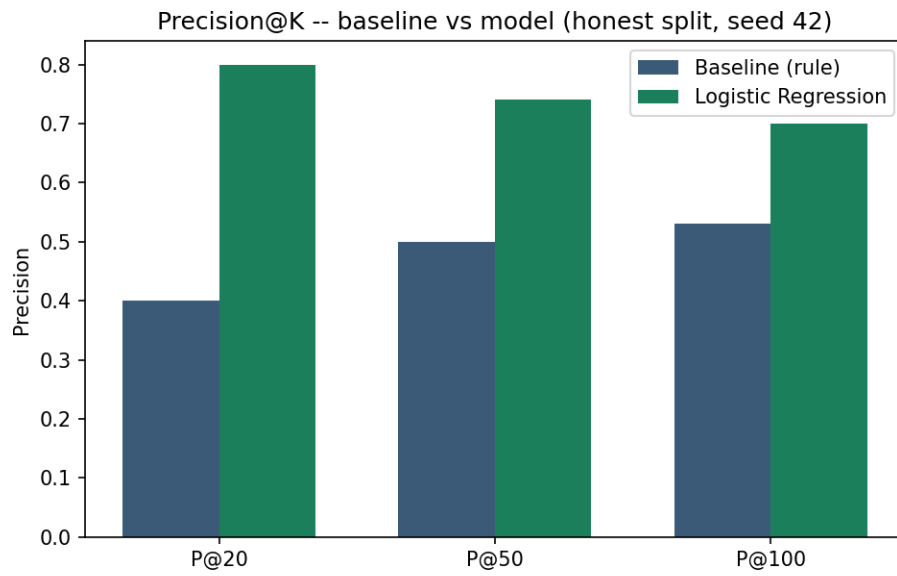


Figure 1: Precision@K, baseline vs. model, same honest split (seed 42).

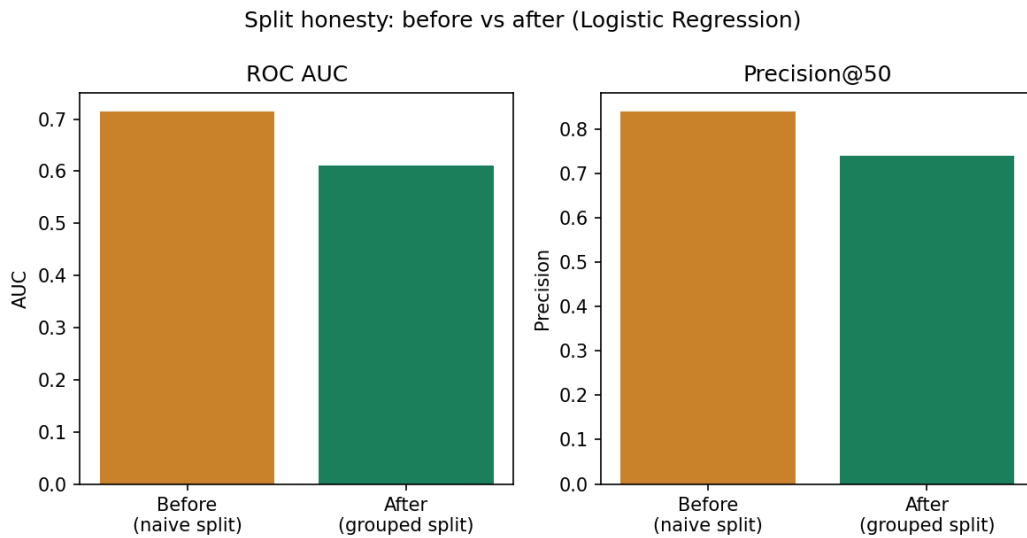


Figure 2: Before (naive row split) vs. after (client-grouped split): ROC AUC 0.72 $\rightarrow$ 0.61, Precision@50 0.84 $\rightarrow$ 0.74.

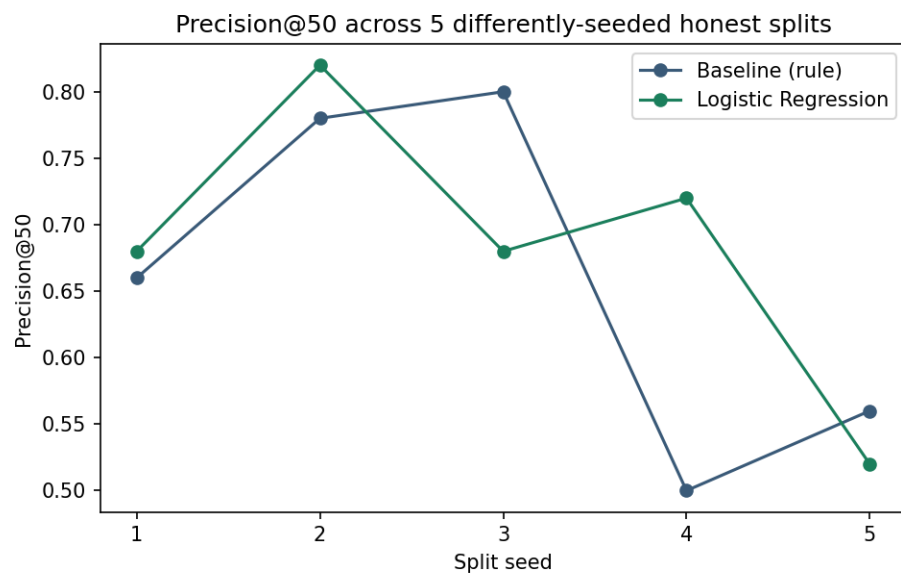


Figure 3: Precision@50 across 5 differently-seeded honest splits. Logistic Regression led in 3 of 5.

5 Limitations & Honest Framing

Absolute-gap scoring structurally favors the `page_1` tier over `striking/page_3_5/deep`, even when relative underperformance is larger elsewhere.

One client can dominate the top of the ranked queue — there is no per-client normalization in the current rule.

`top_3`'s own expected-CTR benchmark returned a MIXED signal check, not CONFIRMED — treat top-tier flags with extra caution.

The model beat the rule in 3 of 5 differently-seeded splits, not 5 of 5. It is supporting evidence, not a replacement for the rule.

Most model signal comes from features overlapping the label's own window, not durable content properties — read model output as “resembles other declining pages,” not “will decline.”

The decline label is a 30-vs-30-day internal comparison, not an externally confirmed outcome.

Only 32 clients total — any single held-out precision number carries real sampling variance; treat it as a range, not a guarantee.

One anonymized data slice, one lane, one time window. Nothing here has been checked against a different lane, period, or unanonymized production data.

6 Ranked Recommendations

The deployed output ranks on the transparent rule, not the model, per the finding in Section 4. Every scored page falls into one of five archetypes, each with a default action a human still has to approve.

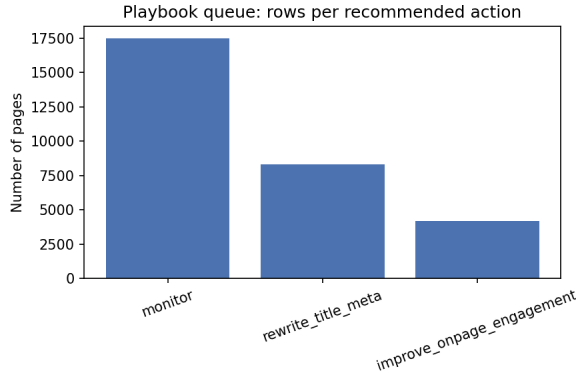
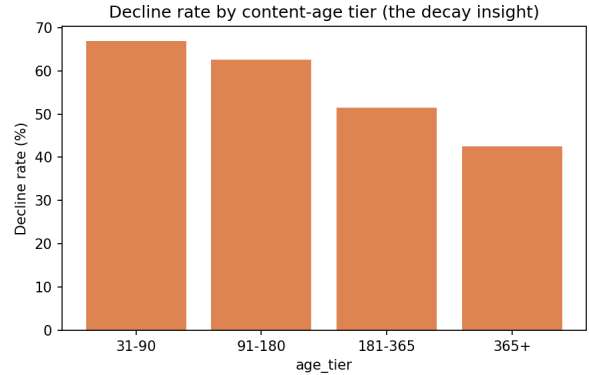
Table 3: Archetype distribution across all 30,000 scored pages.

Archetype	Default action	Pages
Visible & Underclicked	Rewrite title/meta	6,140
Buried & Underperforming	Review before acting	2,198
Ranking Well, Low Engagement	On-page / content re-view	4,175
On Track	No action	6,239
Not Enough Data Yet	Monitor only	11,248

Table 4: Top 10 of the ranked queue. Content IDs are anonymized hashes.

Rank	Archetype	Position tier	Score
1	Visible & Underclicked	page_1	0.946
2	Ranking Well, Low Engagement	page_1	0.939
3	Visible & Underclicked	page_1	0.936
4	Visible & Underclicked	page_1	0.932
5	Visible & Underclicked	page_1	0.932
6	Visible & Underclicked	page_1	0.931
7	Visible & Underclicked	page_1	0.922
8	Visible & Underclicked	page_1	0.918
9	Ranking Well, Low Engagement	page_1	0.910
10	Visible & Underclicked	page_1	0.910

Note the top 10 skew entirely to *page_1* — a direct, visible instance of the scoring-bias limitation above, left uncorrected in this baseline on purpose (see Section 5).

**Figure 4:** Full queue by recommended action.**Figure 5:** Decline rate *falls* with content age.

What should not be automated: no auto-publishing of any rewrite; no auto-deprioritizing a client's pages based on score alone; no hard cutoff on the model's probability for automated action; no reuse of this scoring for a different lane or client data without re-running this validation; no treating the decline label as a confirmed outcome in downstream reporting. Full human-review rules, cost/value notes, and monitoring triggers are in the accompanying playbook notebook.

7 Reproducibility

Every number and figure in this report traces back to a fixed seed (42) and a committed results file (`work/outputs/capstone_results.json`). Anyone with the repository can re-run the full chain end to end.

- Repository: <https://github.com/Rida-create/flyrank-ml-internship-starter>
- `work/notebooks/w04_baseline_score.ipynb` — baseline rule + signal checks
- `work/notebooks/w05_model.ipynb` — model vs. baseline
- `work/notebooks/w06_validation_audit.ipynb` — validation audit (split honesty, leakage)
- `work/notebooks/w07_action_playbook.ipynb` — action playbook
- `work/notebooks/capstone.ipynb` — this report’s source of record

Each notebook is Colab-ready: it clones the repository and installs dependencies automatically if it doesn’t detect a local checkout. Locally, `python scripts/run_all.py` from the repository root runs the original reference pipeline on the bundled sample. This $\text{\LaTeX}$ source (`paper/paper.tex`) compiles standalone with `pdflatex` and the figures bundled in `paper/figures/`.

Acknowledgments & Data Credit

This work was completed as part of the FlyRank ML Internship — **Applied Search Intelligence: Google Search Ranking & Discoverability** — and relies throughout on the anonymized sample data, framework documents, and reference pipeline the program provides.

Built on the **FlyRank ML Internship** dataset.

<https://flyrank.ai>

All data in this report is anonymized per the program’s data-use policy: no client names, domains, URLs, or query text. Findings are observational and directional, intended as decision support for a human review process — not a causal claim about search ranking, and not an autonomous system.